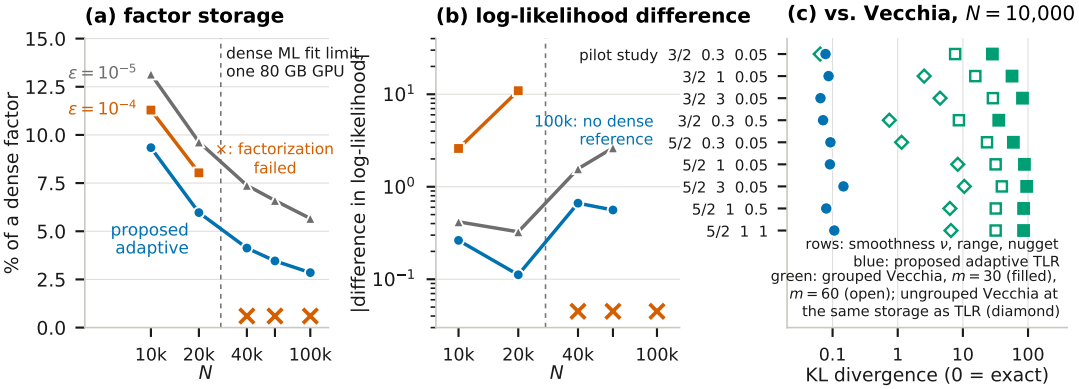
(a) factor storage
% of a dense factor
15.0
12.5
10.0
7.5
5.0
2.5
0.0
$\epsilon = 10^{-5}$
$\epsilon = 10^{-4}$
dense ML fit limit,
one 80 GB GPU
×: factorization
failed
proposed
adaptive
10k
20k
40k
100k
N
(b) log-likelihood difference
|difference in log-likelihood|
10^{1}
10^{0}
10^{-1}
100k: no dense
reference
10k
20k
40k
100k
N
(c) vs. Vecchia, $N = 10{,}000$
pilot study 3/2 0.3 0.05
3/2 1 0.05
3/2 3 0.05
3/2 0.3 0.5
5/2 0.3 0.05
5/2 1 0.05
5/2 3 0.05
5/2 1 0.5
5/2 1 1
rows: smoothness ν, range, nugget
blue: proposed adaptive TLR
green: grouped Vecchia, m = 30 (filled),
m = 60 (open); ungrouped Vecchia at
the same storage as TLR (diamond)
0.1
1
10
100
KL divergence (0 = exact)